Shiyuan Zhang

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Education

08/2023–Current **Master of Computer Science**
University of Illinois at Urbana-Champaign
Cumulative GPA: 4.00

08/2019–05/2023 **B.S Statistics**
Minor in Computer Science
University of Illinois at Urbana-Champaign
Cumulative GPA: 3.96

Previously enrolled in the joint Information & Computer Science Bachelor's program at University of Liverpool & XJTLU (non-degree coursework)

Research Interest

ML Explainability: Data Attribution Method on Generative Model
Trustworthy ML: Model Fairness, Model Robustness, Model safety
NLP: Societal Effects and Social Bias of Language Models

Publications

1. Deng*, J., Li*, T. W., **Zhang, Shiyuan**, Liu, S., Pan, Y., Huang, H., Wang, X., Hu, P., Zhang, X., and Ma, J. *dattri*: A Library for Efficient Data Attribution. *Submitted to NeurIPS 2024 Datasets and Benchmarks Track*
2. Ge*, Y., **Zhang, Shiyuan***, and Liu, X. Nuanced Multi-class Detection of Machine-Generated Scientific Text. *Fourth Workshop on Scholarly Document Processing at ACL 2024*
3. Deng, J., **Zhang, Shiyuan**, and Ma, J. Computational Copyright: Towards A Royalty Model for AI Music Generation Platforms. *ISMIR 2024*
(* Equal Contribution)

Research Experience

02/2024–Current **Trustworthy and Regulatable AI Systems Lab, UIUC**
Mentor: Professor Jiaqi Ma

1. **Development of the Open-Source Library *dattri***

- Implemented core functions, including Leave-One-Out (LOO), Linear Datamodeling Score (LDS), and Inverse Hessian Vector Product (IHVP).
- Designed and implemented a unified API, simplifying the switch between different data attribution methods.
- Integrated nanoGPT trained on different datasets, MNIST+MLP, and MNIST+LR, into the benchmark framework to compare the performance and effectiveness of different data attribution methods.

2. **Data Attribution Method for Generative Music Models**

- Implemented TRAK and TracIn frameworks to calculate the influence score of training data points on Music Transformer models.
- Evaluated the effectiveness of data attribution models using linear data modeling scores.

- Designed an adversarial attack using repetition and copy-paste techniques on the music dataset to test the robustness of the data attribution model.

12/2023–05/2024 **Independent Research, UIUC**
Collaborator: Yubin Ge
Enhanced Text Classification with Contrastive Learning

- Collected and filterd text data using the Semantic Scholar API and generated text by using language models (GPT-3.5) to simulate generating methods such as paraphrasing, summarization, and title-based generation.
- Proposed and implemented a novel baseline model using contrastive learning based on SciBert, RoBerta, and DeBerta, significantly improving text classification accuracy and robustness.

Teaching Experience

05/2024–Current **CEE 202 - Eng Risk & Uncertainty, UIUC**
Teaching Assistant Professor Sotiria Koloutsou-Vakakis

- Conducted weekly office hours to support students programming assignments and questions about the lecture content.
- Led R programming tutorials, graded assignments, and responded to student inquiries on Campuswire.

09/2023–06/2024 **SIIP Program for CEE 202, CEE 330, CEE 340, UIUC**
Teaching Assistant Professor Sotiria Koloutsou-Vakakis

- Developed automated grading systems on the PrairieLearn and Gradescope platform(Python).
- Designed Jupyter Notebook assignments, and created R/Python problem sets and corresponding answer sets.

05/2022–08/2022 **CS 411 - Database Systems, UIUC**
Course Assistant Professor Abdussalam Alawini

- Proctored exams, graded assignments and provided homework assistance.
- Designed and checked SQL questions on PrairieLearn platform.

Skills

Programming	Python, C/C++, R
Tools	Pytorch, MySQL, MongoDB, Neo4j, Linux, Git, Docker, Notebook, Slurm, Ubuntu

Honors, Awards & Scholarships

08/2019-05/2023	UIUC Dean's List in all the semesters
05/2023	Graduated with Highest Distinction

Selected courses

ML	Advanced NLP, Applied ML, AI Algorithms, Computer Vision
System	Software Engineering, Database, Computer Network, Operating System
Stat&Math	Abstract Linear Algebra, Statistical Modeling, Adv Prob&Stat, Math Analysis